MILITARY SPECIFICATION

BATTERY, THERMAL, M17

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of battery, entitled Battery, Thermal, M17 which is used with the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-1169	- Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering Electrical Connections for Special Weapons Systems, Procedure for.

DRAWINGS

PICATINNY ARSENAL

8876382 - Battery Assembly
EL-8876382 - Equipment List

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The battery shall comply with all requirements specified on the drawing (dwg) 8876382, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Initiator circuit resistance. Initiator circuit resistance between the terminals in Table I shall measure 4.6 plus or minus (±) 0.6 ohms when tested as specified in 4.4.1.

Table I

Terminals	Test Code No. (see 3.5)
U and M	050001
N and P	050002

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3.7 Circuit resistance.

3.7.1 Prior to battery activation. Circuit resistance between the terminals shall be as specified in Table II when tested as specified in 4.4.2.1.

Table II

Terminals	Resistance	Test Code No. (see 3.5)
U to M	4.6 ± 0.6	061001
N to P	4.6 ± 0.6	061002
R to S	0.15 ohms (max)	061003
H to J	0.05 " "	061004
B to C	0.05 " "	061005
K to T	0.05 " "	061006
D to E	0.05 " "	061007
A to L	0.05 " "	061008
F to G	0.05 " "	061009

3.7.2 After battery initiation. The monitor circuit resistance between terminals R and S shall be 1.0 megohms minimum after battery has cooled to ambient temperature when tested as specified in 4.4.2.2.

3.8 Polarity. The terminals when connected in the combinations listed in Table III shall indicate a positive (+) voltage while being tested as specified in 4.4.3.

Table III

Positive Terminals		Negative Terminals
A or L	to	F or G
H or J	to	B or C
K or T	to	D or E

3.9 Insulation resistance. The insulation resistance between the terminals jointly grouped as indicated in Table IV and the battery case shall measure 100 megohms, minimum, at a temperature of 160 degrees ± 5 degrees Fahrenheit (°F). Testing shall be in accordance with 4.4.4.

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Table IV

Terminals				
Match U and M	to all others and case			
Match N and P	"			
Monitor R or S	"		except unused	monitor term.
Sec 1 + H or J	"	except	unused	Sec 1 + terminal
Sec 1 - B or C	"	"	"	Sec 1 - "
Sec 2 + K or T	"	"	"	Sec 2 + "
Sec 2 - D or E	"	"	"	Sec 2 - "
Sec 3 + A or L	"	"	"	Sec 3 + "
Sec 3 - F or G	"	"	"	Sec 3 - "

3.10 Gross leak. There shall be no emission of bubbles while the battery is submerged in water while being tested as specified in 4.4.5.

3.11 Activation current. The battery shall activate when each initiator is subjected to a current of 1 ± 0.1 amperes direct current (dc). Testing shall be as specified in 4.4.6.

3.12 Electrical performance. During activated life, the battery shall produce the minimum voltage and current for each event in Table V at the magnitudes described, after being stabilized at -40 or 160°F as applicable, when tested in accordance with 4.4.7. After testing as specified in 4.4.7, the battery shall be tested in accordance with 4.4.2 for compliance with 3.7.2.

Table V

Load Description

Event	Interval	Resistance Ohms	Min Req Current Amps	Test Code No. See 3.5
	Section I			
A	0 to 10 Volts	1.5	6	110001
B	15 V to 28 Volts	Open Ckt	0	110002
C	28 V to 7.5 Secs	Open Ckt	0	110003
D	7.5 to 7.6 Secs	11.9	2.35	110004
E	7.6 to 185 Secs	80	0.350	110005
F	185 to 185.1 Secs	3.25	8.0	110006
G	185.1 to 195 Secs	9.3	3.0	110007
	Section II			
H	From initiate time to 6.75 V	9.3	0.725	110008
	Section III			
J	From initiate time to 6.75	9.3	0.725	110009

NOTE: 1. Zero time corresponds to start of battery match initiate time.

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NOTE: 2. The value of the time at the start of each event except event 1 shall be within + 1.0 second of the initiate time of each event. Load duration shall be no less than the interval described.

3. Resistive loads only.

4. The voltage shall be between 28 and 33 volts when either maximum or minimum current is supplied, except that open circuit voltage may be 35 volts maximum. The voltage of the 8 volt sections shall be between 6.75 and 8.10 volts. The voltage at Event F is permitted to drop to 26.0 volts minimum.

5. Section I consists of a nominal 30 volt section. Sections II and III are each nominal 8 volt sections.

6. Event A - Record time to reach 15 volts.

7. Event B - Record time to reach 28 volts.

8. Event F - Record voltage of Section I.

9. Events H & J - Record time to drop to 6.75 volts.

10. Record voltages of Sections I, II, and III at 200 seconds after initiation.

11. Fluctuations for the output voltage for Section I subsequent to Event F shall not exceed 1.0 v peak to peak within frequency range of 80-2000 Hz. Fluctuations in the output voltages for Sections II and III shall not exceed 250 mv peak to peak within frequency range of 80-2000 Hz.

12. The requirements of Note 4 above shall be met for a minimum of 200 seconds.

13. The rise time to reach 28 volts for Section I, and Sections II and III to reach 6.75 volts shall be less than 3 seconds.

14. Event G - Record time to drop to 28v.

15. Resistance values shall be within 5% - 0% as shown in the table.

16. Voltage readings of Sections I, II and III will be made and recorded at 60, 120, 180, and 200 seconds as specified in 4.4.7.

3.13 Visual indicator. The visual indicator of the battery shall undergo a permanent color change from pink to blue within ten minutes after activation.

3.14 Temperature humidity. The battery shall be exposed to temperature humidity for fourteen days with temperatures ranging from -65 to +160°F. At the completion of the final cycle, the battery shall meet with the requirements of 3.9. Testing shall be in accordance with 4.4.12.

3.15 Transportation vibration. The battery shall withstand the vibration conditions specified in Table VI along each of its three mutually perpendicular axes (see Figure 1). Following this test as specified in 4.4.13, the battery shall be tested for compliance with 3.6, 3.7.1, 3.8, 3.9 and 3.10 after being stabilized (see 6.4) at ambient temperature. Frequency shall be cycled at a logarithmic rate between 5-500-5 Hertz (Hz) in 7.5 minutes.

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Table VI

Frequency Hz	Gravity Units (g's) or Amplitude (da-in.)	Time Minutes Per Axis -65°+5°F	Time Minutes Per Axis +160°+5°F
5-14-5	0.4 g	15	15
15-74-15	0.036 double amplitude	15	15
75-500-75	10.0 g	15	15

3.16 Transportation shock. The battery shall withstand a shock of 60 ± 6 gravity units (g) along each of its three mutually perpendicular axes (see Figure 1). Each shock along each axis shall be half-sine wave shape and shall be 11 ± 1 millisecond (ms) in duration. The time to peak shall be 5 ±1 ms. There shall be a total of eighteen shocks, three shocks in each direction in each of three axes, after being stabilized, at -65 or +160°F, as applicable. At the completion of these tests, the battery shall be stabilized at ambient temperature and then tested for compliance with 3.6, 3.7.1, 3.8, 3.9 and 3.10. Testing shall be as specified in 4.4.14.

3.17 High temperature. The battery shall withstand a temperature of 160±5°F or 140±5°F as applicable for a min of four hours or until stabilization (see 6.4). After stabilization, the battery shall be tested within five minutes in accordance with the applicable test paragraph of 4.4. At the completion of these tests (except transverse acceleration-temperature 4.4.11), the battery shall comply with 3.6, 3.6.1, 3.8, 3.9 and 3.10 when tested as specified in 4.4.8.

3.18 Low temperature. The battery shall withstand a temperature of -65±5°F or -40±5°F as applicable for a min of four hours or until stabilization (see 6.4). After stabilization, the battery shall be tested within five minutes in accordance with the applicable test paragraph of 4.4. At the completion of these tests (except transverse acceleration-temperature 4.4.11), the battery shall be returned to ambient temperature and comply with 3.6, 3.7.1, 3.8, 3.9 and 3.10 when tested as specified in 4.4.9.

3.19 Operation vibration. The battery shall be vibrated in each of three mutually perpendicular axes (see Figure 1), after being stabilized at -65 or 160°F as applicable. The inputs shall be applied to the following sequence:

0.3 g^2/Hz power spectral density (19.0 root means square (rms) g's min) at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibel (db)/octave so that the total frequency envelope is 50 to 2000 Hz for 10 seconds. Tolerance level: ± 3 db between frequencies of 100 and 1000 Hz and ±

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6.0 db between frequencies of 1000-2000 Hz.

0.06 g^2/Hz power spectral density (8.5 rms g's min) at frequencies ranging from 100 to 1000 Hz, attenuated 12 db/ octave so that the frequency envelope is 50 to 2000 Hz for one minute. Tolerance level: ± 3 db between frequencies of 100 - 2000 Hz.

After completion of operation vibration test, the battery shall be returned to ambient temperature and shall then meet the initial requirements of 3.6, 3.7.1, 3.8 and 3.10, and then of 3.11 and 3.12. The test shall be conducted as specified in 4.4.10.

3.20 Transverse acceleration-temperature. The battery shall withstand an acceleration force of 15 ± 1 g's as shown in Figure 1. The radius arm shall be computed to the center of the cylindrical section of the battery.

15 ± 1 g's acceleration shall be applied continually from initiation of the battery until Figure 1 and Table V conditions have been satisfied. Before the battery is accelerated, it shall be stabilized at 160 or -40°F as applicable. During this acceleration test, the battery shall comply with all requirements of 3.11, 3.12; and after testing, shall comply with 3.13.

3.21 Reliability. The battery reliability shall be a minimum of 0.99. Testing shall be as specified in 4.3.3.3.

3.22 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.23 Workmanship.

3.23.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.23.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.23.3 Burrs. Parts shall be free from burrs.

3.23.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.23.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 16 complete assemblies (see dwg 8876382) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VII. Tests shall be conducted in order specified in Table VII.

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Table VII
First article inspection and tests

Inspection or Test	Reqmt Para	Battery No. 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
a. Classification of defects/ characteristics	4.3.2.1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
b. Gross leak	3.10	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
c. Polarity	3.8	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
d. Initiator circuit res.	3.6	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
e. Fuze circuit resistance	3.7.1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
f. Insulation resistance	3.9	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
g. Temperature-humidity (see Note 3)	3.14	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
h. Transportation vibration (see Note 2)	3.15																
(1) High temperature (160°F)	3.17	X	X				X	X			X	X		X			X
(2) Low temperature (-65°F)	3.18			X	X	X			X	X			X		X	X	
i. Transportation shock (see Note 2)	3.16																
(1) High temperature (160°F)	3.17	X		X			X		X		X		X		X		X
(2) Low temperature (-65°F)	3.18		X		X	X		X		X		X		X		X	
j. Operation vibration (see Note 2)	3.19																
(1) High temperature (160°F)	3.17	X			X			X	X			X	X			X	X
(2) Low temperature (-65°F)	3.18		X	X		X	X			X	X			X	X		
k. Transverse acceleration (15 g) (see Note 1)	3.20																
(1) High temperature (140°F)	3.17	X	X	X	X	X	X	X	X								
(2) Low temperature (-40°F)	3.18									X	X	X	X	X	X	X	X

Note 1. During the environmental test k, the batteries shall be activated and shall comply with the requirements of 3.11 and 3.12.
Note 2. These tests are conditioning tests. The batteries will not be activated.
Note 3. At the completion of the temperature humidity test, check all batteries in accordance with 3.6, 3.7.1, 3.8, 3.9 and 3.10.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Production inspection examination. All batteries in the inspection lot shall be tested 100 percent for the characteristics specified in 4.3.2.1.1. Any characteristic found defective shall reject that battery from the lot.

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 M17 Battery (See dwg. 8876382).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect length, less terminals (max)	Gage
102.	Incorrect outside diameter (max)	Gage
103.	Mislocation of mounting holes	Gage
104.	Incorrect diameter of mounting holes	Gage
105.	Case damaged	Visual
106.	Plating finish improper	Visual
107.	Terminal glass chipped	Visual
108.	Identification plate, missing, misleading or unidentifiable	Visual
109.	Evidence of poor workmanship (see 3.23)	Visual
110.	Temperature indicating paint missing	Visual
111.	Inability of connectors to mate and unmate without the use of special tools	Gage

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Categories	Defects/Characteristics	Method of Inspection
Major: (Con't)		
	112. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor: None defined.		

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
	101. Preservation, packaging and packing not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.23)	Visual

4.3.3 Inspection Testing. The contractor shall perform all tests, and in the order shown, in Table VIII.

Inspection Testing

Table VIII

Tests	Reqmt. Para.	Number Of Test Items	Remarks
a. Gross Leak	3.10	100% of Lot	See 4.3.3.1
b. Polarity	3.8	100% of Lot	See 4.3.3.1
c. Initiator circuit res.	3.6	100% of Lot	See 4.3.3.1
d. Monitor circuit res.	3.7	100% of Lot	See 4.3.3.1
e. Insulation resistance	3.9	100% of Lot	See 4.3.3.1
f. Transportation shock	3.16		See 4.3.3.2 & Note 2 below
@ High Temp (+160°F)	3.17	Test Sample (16)	See 4.3.3.2 & Note 2 below

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	@ Low Temp (-65°F)	3.18	Test Sample (16)	See 4.3.3.2 & Note 2 below
g.	Transportation vibration	3.15	Test Sample (16)	See 4.3.3.2 & Note 2 below
h.	Tranverse acceleration	3.20		See 4.3.3.3 & Notes 1 & 2 below
	@ High Temp (+140°F)	3.17	Test Sample (8)	See 4.3.3.3 & Notes 1 & 2 below
	@ Low Temp (-40°F)	3.18	Test Sample (8)	See 4.3.3.3 & Notes 1 & 2 below

Note 1: Test h is destructive and samples so tested shall not be returned to the lot.

Note 2: In tests f, g, h, if any requirement is not satisified the lot shall be rejected.

*Note 3: All of the above shall have a test classification of Major.

4.3.3.1 Leakage (3.10), polarity (3.8), initiator circuit resistance (3.6), fuse circuit resistance (3.7.1), insulation resistance (3.9). Each battery in lot shall be subjected to these tests. If any battery fails to comply with any requirement, when tested as specified in the applicable paragraphs under 4.4.5, 4.4.3, 4.4.1, 4.4.2.1, and 4.4.4. The tests shall be classified defective, and removed from the lot.

4.3.3.2 Transportation shock (3.17) and transportation vibration (3.16). A test sample of 16 batteries shall be selected at random from each inspection lot of 300 or portion thereof. Each of the sixteen batteries shall be subjected to these tests.

4.3.3.3 Transverse acceleration - temperature (3.20). Eight (8) of the sixteen (16) batteries shall be subjected to high temperature per 3.17 and per 3.20. The eight (8) remaining batteries shall be subjected to low temperature per 3.18 and functioned per 3.20. Variable data will be taken per 3.11 and 3.12. The mean and standard deviation of the voltage at minimum life shall be calculated for both groups of eight (8) batteries. The K factor for each group shall be calculated as follows:

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K=(X-R)/S where X is the mean of the sample: R is the required voltage at the minimum life time and S is the standard deviation of the sample about its mean. The K factor shall have a minimum value of 2.35 corresponding to a minimum reliability of of 0.99. If the sample fails to meet the minimum reliability, the lot shall be rejected. If any battery fails to comply with any requirement, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8876382). When either of the conditions listed below exist the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the test equipment used. All tests are conducted at room ambient temperature (80 ± 20°F) unless specified otherwise.

CAUTION: Prior to conducting the insulation resistance test, all battery circuits (including initiator, fuse, battery, circuit to circuit, and all circuits to case) shall be checked for shorts and low resistance. Under no conditions may an ohmmeter or other instrument without current limited capability be placed across any battery terminal.

4.4.1 Initiator circuit resistance. The circuit resistance of each initiator shall be measured by utilizing a suitable circuit resistance tester with current limited to 10 milliamperes. The measured resistance shall be as specified in 3.6. The ambient temperature shall be 77 ± 8°F.

4.4.2 Other circuit resistance tests.

4.4.2.1 Prior to battery initiation. The fuse circuit resistance shall be measured before battery initiation by utilizing a suitable circuit resistance tester with current limited to 10 milliamperes for points U to M and N to P and 100 milliamperes for the rest of the points. The measured resistance shall be as specified in 3.7.1.

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4.4.2.2 After battery initiation. The monitor circuit resistance shall be measured after battery initiation by a suitable resistance measuring device, using 110 VDC max and 5 milliamperes max. The measured resistance shall be as specified in 3.7.2.

4.4.3 Polarity. The battery shall be tested for proper polarity with an electrometer (input impedance should be at least 10^{12} ohms) to determine compliance with 3.8. The battery shall be heated in an oven at 160 + 5°F and maintained at this temperature for one hour. Within five minutes of this period, the battery shall be tested.

4.4.4 Insulation resistance. The battery shall be heated in an oven at 160 + 5°F and maintained at this temperature for one hour. The insulation resistance between terminals as specified in Table IV shall be measured for a minimum of five seconds or until megohmmeter indicator is stabilized by a suitable insulation resistance measuring device (100 + 10 vdc megohmmeter) with current limited to 5 milliamperes. The measured resistance shall be as specified in 3.9. Measurement shall be completed within 10 minutes of removal from heated chamber.

4.4.5 Gross leak. The sealed battery shall be placed into a chamber of dry nitrogen and pressurized to 90 pounds per square inch (psi) min for 1 hour. The sealed battery shall then be submerged, within 2 minutes after removal from chamber, for 2 minutes min, 5 minutes max, in water. The water shall be maintained at a temperature ranging between 160 and 175°F. Leakage shall be as specified in 3.10.

4.4.6 Activation current. The dc current, as specified in 3.11, shall be applied with a suitable current producing device, of not over 30 v for a period of 20 + 10 milliseconds (ms), and monitored with approved measuring equipment.

4.4.7 Load, life and voltage. The electrical characteristics and time durations shall be measured with a suitable recording instrument, which shall have a galvanometer with a flat response + 5% up to a frequency of 2000 Hz, for voltage measurements, and which gives a permanent record to determine compliance with 3.12 (Note 2). Voltage readings of Sections I, II and III will be made at 60, 120, 180 and 200 seconds + 5 seconds with an approved measuring device of 0.1% minimum accuracy.

4.4.8 High temperature. The battery shall be placed in a high temperature chamber and the temperature maintained at 160+ 5°F or 140 + 5°F as applicable for a min of four hours or until the battery is stabilized at 160°F (see 6.4). After stabilization, the battery shall be removed from the chamber and tested for compliance with requirements as specified in Table VII or Table VIII as applicable. Requirements in accordance with 3.17.

4.4.9 Low temperature. The battery shall be placed in a low temperature chamber and the temperature shall be lowered to and maintained at -65 + 5°F or -40 + 5°F as applicable for a min of four hours or until the battery is stabilized (see 6.4). After stabilization, the battery shall be removed from the chamber and tested for compliance with requirements as specified in Table VII or Table VIII, as applicable.

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4.4.10 Operational vibration. The battery shall be mounted securely to a suitable fixture and vibrated at the levels specified in 3.19. After undergoing vibration, the battery shall meet the post-test requirements of 3.19 and then shall be tested as specified in 4.4.6 and 4.4.7.

4.4.11 Transverse acceleration-temperature. The battery shall be mounted securely to a suitable fixture and then mounted to a suitable spinning apparatus and spun at the specified level after being conditioned per 3.20. While being spun, the battery shall be tested as specified in 4.4.6 and 4.4.7.

4.4.12 Temperature humidity. The battery shall be placed in a suitable temperature humidity chamber and tested in accordance with MIL-STD-331 Test Method 105. At the completion of the tests as specified in 3.14, the battery shall be removed from the chamber and the terminals air dried and tested to meet the requirements of 3.9.

4.4.13 Transportation vibration. The battery shall be mounted securely to a suitable fixture and vibrated at the levels specified in Table VI. After undergoing the transportation vibration, the battery shall meet the post-test requirements of 3.15.

4.4.14 Transportation shock. The battery shall be mounted on a suitable fixture and shocked at the levels and conditions specified in 3.16. After undergoing the transportation shock, the battery shall meet the post-test requirements of 3.16.

5. PREPARATION FOR DELIVERY

5.1 Packing and packaging. The battery shall be packed and packaged in accordance with MIL-STD-1169 level "C".

5.2 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

6.1.2 Contract data requirements (see 3.5). Data specified in 3.6, 3.7.1 and 3.12 will be delivered as identified on a numbered DD Form 1664.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2013 dated 21 April 1971, with Amendment 7 dated 13 February 1975.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A069

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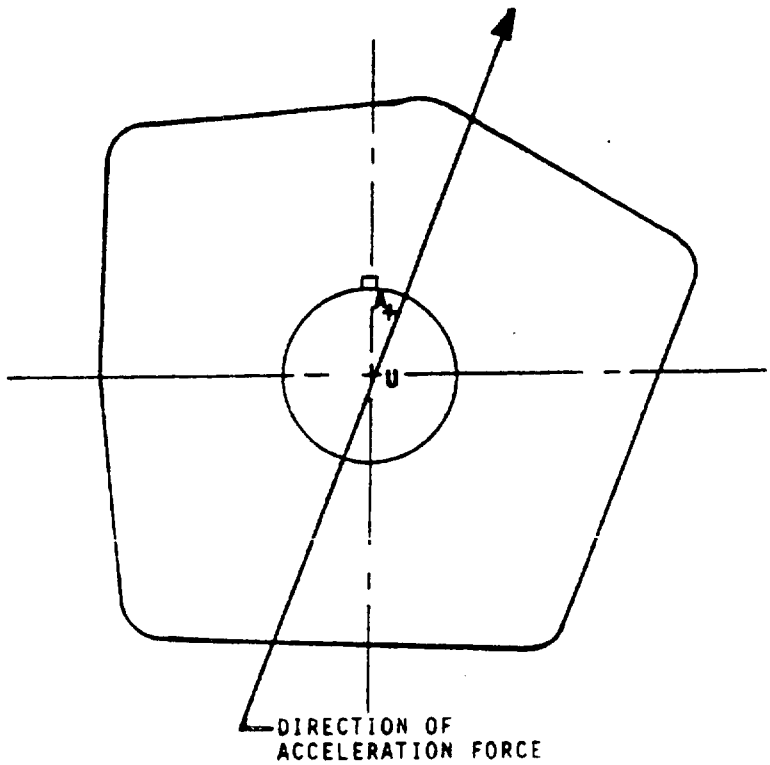


Figure 1. Direction Of Acceleration Force

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POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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